**1** Scale factor $= \dfrac{6+3}{6} = 1.5$ (or ratio $\dfrac{QR}{8} = \dfrac{9}{6}$) **[M1]**

12 **[A1]**

*[Total Marks: 2]*

**2** **(a)** Linear scale factor $= \dfrac{4+2}{4} = 1.5$ (or equivalent ratio $\dfrac{6}{4}$) **[M1]**

7.5 **[A1]**

**(b)** Area of $ABC = 8 \times 1.5^2 = 18$ **[M1]**

10 **[A1]**

*[Total Marks: 4]*

**3** **(a)** Scale factor $= \dfrac{18}{6} = 3$ **[M1]**

15 **[A1]**

**(b)** Area scale factor $= 3^2 = 9$ **[M1]**

$\dfrac{72}{9} = 8$ **[A1]**

*[Total Marks: 4]*

**4** **(a)** Volume scale factor $= \left(\dfrac{15}{6}\right)^3 = \dfrac{125}{8}$ **[M1]**

$V_B = 48 \times \dfrac{125}{8} = 750$ **[A1]**

**(b)** Area scale factor $= \left(\dfrac{15}{6}\right)^2 = \dfrac{25}{4}$ **[M1]**

$A_A = 200 \div \dfrac{25}{4} = 32$ **[A1]**

*[Total Marks: 4]*

**5** **(a)** Length scale factor $= \dfrac{9}{3} = 3$ **[M1]**

15 **[A1]**

**(b)** Area scale factor $= 3^2 = 9$ **[M1]**

6 **[A1]**

*[Total Marks: 4]*

**6** **(a)** Scale factor $= \dfrac{4+2}{4} = 1.5$ (or $\dfrac{6}{4}$ or $\dfrac{3}{2}$) **[M1]**

7.5 **[A1]**

**(b)** Area scale factor $= 1.5^2 = 2.25$ **[M1]**

Area of triangle $ABC = 8 \times 2.25 = 18$ **[M1]**

10 **[A1]**

*[Total Marks: 5]*

**7 (a)** Scale factor $= \dfrac{4+2}{4}$ or 1.5 **[M1]**

9 **[A1]**

**(b)** Area scale factor $= 1.5^2$ or 2.25 **[M1]**

Area of triangle $ABC = 32 \times 2.25$ or 72 **[M1]**

40 **[A1]**

*[Total Marks: 5]*

**8 (a)** Identifies the scale factor is $\dfrac{AD+DB}{AD} = \dfrac{6}{4}$ or 1.5 **[M1]**

9 **[A1]**

**(b)** Uses the area scale factor: $1.5^2$ or $\dfrac{9}{4}$ or 2.25 **[M1]**

Calculates the area of triangle $ABC$: $12 \times 2.25 = 27$ **[M1]**

15 **[A1]**

*[Total Marks: 5]*

**9 (a)** Finds the linear scale factor: $\dfrac{15}{10} = 1.5$ (or $\dfrac{3}{2}$) **[M1]**

Finds the area scale factor: $1.5^2 = 2.25$ (or $\dfrac{9}{4}$) **[M1]**

$120 \times 2.25 = 270$ **[A1]**

**(b)** Finds the volume scale factor: $1.5^3 = 3.375$ (or $\dfrac{27}{8}$) **[M1]**

Sets up the correct division: $540 \div 3.375$ (or $540 \times \dfrac{8}{27}$) **[M1]**

160 **[A1]**

*[Total Marks: 6]*

**10 (a)** $\dfrac{6}{9} = \dfrac{8}{QR}$ or scale factor $= \dfrac{9}{6}$ (or 1.5) **[M1]**

12 **[A1]**

**(b)** $\dfrac{6}{9} = \dfrac{PT}{12}$ or $PT = 12 \div 1.5$ **[M1]**

8 **[A1]**

**(c)** Area scale factor $= \left(\frac{9}{6}\right)^2$ or $1.5^2$ or $2.25$ **[M1]**

Area of triangle $PQR = 20 \times 2.25 = 45$ **[M1]**

25 **[A1]**

*[Total Marks: 7]*

**11 (a)** Area scale factor $= \frac{45}{20}$ or $\frac{9}{4}$ **[M1]**

Linear scale factor $= \sqrt{\frac{9}{4}} = \frac{3}{2}$ **[M1]**

Volume $B = 16 \times \left(\frac{3}{2}\right)^3 = 16 \times \frac{27}{8} = 54$ **[A1]**

**(b)** Height $A = 12 \div \frac{3}{2}$ or $12 \times \frac{2}{3}$ **[M1]**

8 **[A1]**

*[Total Marks: 5]*

**12 (a)** Volume scale factor $= \frac{1350}{400} = \frac{27}{8}$ (or 3.375) **[M1]**

Linear scale factor $= \sqrt[3]{\frac{27}{8}} = \frac{3}{2}$ (or 1.5) **[M1]**

18 **[A1]**

**(b)** Area scale factor $= \left(\frac{3}{2}\right)^2 = \frac{9}{4}$ (or 2.25) **[M1]**

$\frac{360}{2.25} = 160$ **[A1]**

*[Total Marks: 5]*

**13 (a)** Angle $A$ (or $\angle DAE$ and $\angle CAB$) is common **[B1]**

Angle $ADE$ = Angle $ACB$ (given) **[B1]**

Therefore the third angles are equal ($\angle AED = \angle ABC$) and the triangles are similar (AAA) **[B1]**

**(b)** Sets up correct ratio: $\frac{AD}{AC} = \frac{AE}{AB}$ or $\frac{x}{4+5} = \frac{4}{x+(x+1)}$ **[M1]**

$\frac{x}{9} = \frac{4}{2x+1}$ **[M1]**

$2x^2 + x - 36 = 0$ **[M1]**

$(2x+9)(x-4) = 0$ implies $x = 4$ (ignoring negative root) **[A1]**

**(c)** Area scale factor $= \left(\frac{4}{9}\right)^2 = \frac{16}{81}$ **[M1]**

16 : 65 **[A1]**

*[Total Marks: 9]*

**14** **(a)** Linear scale factor $= 14 \div 4$ or 3.5 **[M1]**

21 **[A1]**

**(b)** Area scale factor $= 3.5^2$ or 12.25 **[M1]**

104.125 **[A1]**

*[Total Marks: 4]*

**15** **(a)** Volume scale factor $= \dfrac{1200}{150} = 8$ **[M1]**

Length scale factor $= \sqrt[3]{8} = 2$ **[M1]**

24 **[A1]**

**(b)** Area scale factor $= 2^2 = 4$ **[M1]**

110 **[A1]**

*[Total Marks: 5]*

**16** **(a)** Volume scale factor $= \dfrac{1080}{320}$ or $\dfrac{27}{8}$ or 3.375 **[M1]**

Linear scale factor $= \sqrt[3]{\dfrac{27}{8}}$ or 1.5 **[M1]**

27 **[A1]**

**(b)** Area scale factor $= 1.5^2$ or 2.25 or $\dfrac{9}{4}$ **[M1]**

Surface area of $A = 630 \div 2.25$ or $630 \times \dfrac{4}{9}$ **[M1]**

280 **[A1]**

*[Total Marks: 6]*

**17** **(a)** Volume scale factor $= \left(\dfrac{30}{12}\right)^3$ or $2.5^3$ or 15.625 **[M1]**

5000 **[A1]**

**(b)** Area scale factor $= \left(\dfrac{30}{12}\right)^2$ or $2.5^2$ or 6.25 **[M1]**

$1125 \div 6.25 = 180$ **[A1]**

**(c)** $4.80 \times \left(\dfrac{30}{12}\right)^3$ or $4.80 \times 15.625$ **[M1]**

75 or 75.00 **[A1]**

*[Total Marks: 6]*

**18** **(a)** Volume scale factor $= \dfrac{8640}{320} = 27$ **[M1]**

Length scale factor $= \sqrt[3]{27} = 3$ **[M1]**

36 **[A1]**

**(b)** Area scale factor $= 3^2 = 9$ **[M1]**

250 **[A1]**

**(c)** Mass $= 8.5 \times 320 = 2720\,\text{g}$ **[M1]**

2.72 **[A1]**

*[Total Marks: 7]*

**19** **(a)** Angle $A$ is common to both triangles **[B1]**

Angle $ABC$ = Angle $ADE$ (corresponding angles as $BC$ is parallel to $DE$) **[B1]**

**(b)** $\dfrac{DE}{7.5} = \dfrac{6+4}{6}$ **[M1]**

12.5 **[A1]**

**(c)** Area scale factor $= \left(\dfrac{10}{6}\right)^2 = \dfrac{25}{9}$ **[M1]**

Area of $ADE = 18 \times \dfrac{25}{9} = 50$ **[M1]**

Area of $BCED = 50 - 18 = 32$ **[A1]**

*[Total Marks: 7]*

**20** **(a)** States that angle $A$ is common to both triangles. **[B1]**

States that angle $ADE$ = angle $ABC$ (and/or angle $AED$ = angle $ACB$) because they are corresponding angles (due to parallel lines). **[B1]**

**(b)** Calculates the full length of $AB$: $6 + 4 = 10$ **[M1]**

Finds the linear scale factor: $\dfrac{10}{6} = \dfrac{5}{3}$ and sets up equation $BC = 9 \times \dfrac{5}{3}$ **[M1]**

15 **[A1]**

**(c)** Finds the area scale factor: $\left(\dfrac{5}{3}\right)^2 = \dfrac{25}{9}$ **[M1]**

Calculates the area of triangle $ABC$: $27 \times \dfrac{25}{9} = 75$ **[M1]**

Calculates the area of the trapezium: $75 - 27 = 48$ **[A1]**

*[Total Marks: 8]*

**21** **(a)** Linear scale factor $= \dfrac{45}{15} = 3$ **[M1]**

Mass (Volume) scale factor $= 3^3 = 27$ **[M1]**

13.5 **[A1]**

**(b)** Mass scale factor from $P$ to $R = \dfrac{4000}{500} = 8$ **[M1]**

Linear scale factor $= \sqrt[3]{8} = 2$ **[M1]**

30 **[A1]**

**(c)** Area scale factor from $P$ to $Q = 3^2 = 9$ **[M1]**

270 **[A1]**

*[Total Marks: 8]*

**22 (a)** Calculates the volume ratio: $\dfrac{1080}{320} = \dfrac{27}{8}$ or 3.375 **[M1]**

Calculates the length scale factor: $\sqrt[3]{\dfrac{27}{8}} = \dfrac{3}{2}$ or 1.5 **[M1]**

18 **[A1]**

**(b)** Uses the area scale factor: $450 \div 1.5^2$ or $450 \times \left(\dfrac{2}{3}\right)^2$ **[M1]**

200 **[A1]**

**(c)** Calculates the volume of cone $R$: $320 + 1080 = 1400$ **[B1]**

Finds the length scale factor from $P$ to $R$: $\sqrt[3]{\dfrac{1400}{320}}$ or $\sqrt[3]{4.375}$ **[M1]**

Calculates the height of $R$: $12 \times \sqrt[3]{4.375} = 19.626\ldots$ and rounds to 19.6 **[A1]**

*[Total Marks: 8]*

**23 (a)** Angle $AXB$ = Angle $CXD$ (vertically opposite) **[B1]**

Angle $ABX$ = Angle $CDX$ (alternate angles, $AB$ parallel to $DC$) **[B1]**

(Accept Angle $BAX$ = Angle $DCX$ for the second B1 mark)

**(b)** Length scale factor $= \dfrac{15}{9} = \dfrac{5}{3}$ **[M1]**

$DX = 6 \times \dfrac{5}{3} = 10$ **[M1]**

$BD = 6 + 10 = 16$ **[A1]**

**(c)** Area scale factor $= \left(\dfrac{15}{9}\right)^2 = \dfrac{25}{9}$ **[M1]**

Area $= 27 \times \dfrac{25}{9} = 75$ **[A1]**

**(d)** Ratio of bases $\dfrac{CX}{AX} = \dfrac{15}{9} = \dfrac{5}{3}$ **[M1]**

Area $= 27 \times \dfrac{5}{3} = 45$ **[A1]**

*[Total Marks: 9]*

**24** **(a)** $\dfrac{4}{4+x} = \dfrac{5}{7+x}$ OR $\dfrac{4}{x} = \dfrac{5}{x+2}$ **[M1]**

$4(x+7) = 5(x+4)$ OR $4(x+2) = 5x$ **[M1]**

$x = 8$ **[A1]**

**(b)** Linear scale factor $= \dfrac{4+8}{4} = 3$ **[M1]**

$BC = 3.5 \times 3 = 10.5$ **[A1]**

**(c)** Area of $\triangle ABC = 3^2 \times A = 9A$ **[M1]**

Area of trapezium $= 9A - A = 8A$ **[A1]**

*[Total Marks: 7]*

**25** **(a)** Volume ratio $A : B = \dfrac{108}{500} = \dfrac{27}{125}$ **[M1]**

Area ratio $= \left(\sqrt[3]{\dfrac{27}{125}}\right)^2 = \left(\dfrac{3}{5}\right)^2 = \dfrac{9}{25}$ **[M1]**

Let surface areas be $9x$ and $25x$. $9x + 25x = 340 \implies 34x = 340 \implies x = 10$ **[M1]**

$25 \times 10 = 250$ **[A1]**

**(b)** Volume of small cone $= 1 - \dfrac{7}{8} = \dfrac{1}{8}$ of Cone $B$ **[M1]**

Ratio of heights $= \sqrt[3]{\dfrac{1}{8}} = \dfrac{1}{2}$ **[M1]**

Height of small cone $= 20 \times \dfrac{1}{2} = 10$ **[A1]**

*[Total Marks: 7]*

**26** **(a)** $\angle AXB = \angle DXC$ (vertically opposite angles) **[B1]**

$\angle BAX = \angle CDX$ OR $\angle ABX = \angle DCX$ (angles in the same segment) **[B1]**

Third angle in triangle OR states the other pair of angles in the same segment, concluding triangles are similar (AAA) **[B1]**

**(b)** Linear scale factor $= \dfrac{7.5}{5} = 1.5$ OR $\dfrac{DX}{4} = \dfrac{7.5}{5}$ **[M1]**

$DX = 6$ **[A1]**

$DC = 9$ **[A1]**

**(c)** Area scale factor $= 1.5^2 = 2.25$ **[M1]**

26.55 **[A1]**

*[Total Marks: 8]*

**27** **(a)** $\dfrac{x+2}{(x+2)+(x-1)} = \dfrac{8}{8+x}$ or $\dfrac{x+2}{x-1} = \dfrac{8}{x}$ **[M1]**

$(x+2)(x+8) = 8(2x+1)$ or $x(x+2) = 8(x-1)$ **[M1]**

$x^2 + 10x + 16 = 16x + 8$ or $x^2 + 2x = 8x - 8$ leading to $x^2 - 6x + 8 = 0$ **[A1]**

**(b)** $(x - 2)(x - 4) = 0$ **[M1]**

2 and 4 **[A1]**

**(c)** When $x = 4$, length ratio $\dfrac{AD}{AB} = \dfrac{6}{9} = \dfrac{2}{3}$ **[M1]**

Area of $ABC = 24 \div \left(\dfrac{2}{3}\right)^2 = 54$ **[M1]**

30 **[A1]**

*[Total Marks: 8]*

**28 (a)** Angle $A$ is common to both triangles **[B1]**

Angle $AED$ = Angle $ABC$ (given in the question) **[B1]**

Therefore, Angle $ADE$ = Angle $ACB$ (sum of angles in a triangle is 180°), so the triangles are similar (AAA) **[B1]**

**(b)** $AB = 6 + 9 = 15$ cm **[M1]**

$\dfrac{AC}{AD} = \dfrac{AB}{AE} \implies \dfrac{AC}{6} = \dfrac{15}{5}$ **[M1]**

$AC = 18$, so $EC = 18 - 5 = 13$ **[A1]**

**(c)** Area scale factor for the triangles $= \left(\dfrac{5}{15}\right)^2 = \dfrac{1}{9}$ **[M1]**

1 : 8 **[A1]**

*[Total Marks: 8]*

**29 (a)** States that angle $BAX$ = angle $DCX$ OR angle $ABX$ = angle $CDX$ due to alternate angles (between parallel lines) **[B1]**

States a second valid pair of angles are equal (e.g. vertically opposite angles at $X$ or the other pair of alternate angles) concluding the triangles are similar (AAA) **[B1]**

**(b)** Identifies the length scale factor: $\dfrac{10.5}{7}$ or 1.5 **[M1]**

Calculates $CX = 4 \times 1.5 = 6$ **[M1]**

10 **[A1]**

**(c)** Calculates the area of triangle $CDX$: $16 \times 1.5^2 = 36$ **[M1]**

Recognises the ratio of areas of triangle $BCX$ to triangle $ABX$ is equal to the ratio of bases $CX : AX = 1.5 : 1$ (as they share the same height from $B$) **[M1]**

Calculates the area of triangle $BCX$ (or $ADX$): $16 \times 1.5 = 24$ **[M1]**

Total area $= 16 + 36 + 24 + 24 = 100$ **[A1]**

*[Total Marks: 9]*

**30 (a)** Angle $A$ is common to both triangles **[B1]**

Angle $ABC$ = Angle $AED$ (given), so the third angles must be equal, hence AA similarity **[B1]**

**(b)** 2.5 or $\frac{5}{2}$ **[B1]**

**(c)** $\frac{AB}{AE} = 2.5 \implies \frac{2x+2.5}{x} = 2.5$ (or equivalent valid ratio) **[M1]**

$2x + 2.5 = 2.5x$ **[M1]**

$x = 5$ **[A1]**

**(d)** Area scale factor $= 2.5^2 = 6.25$ **[M1]**

Formulates correct equation for areas, e.g., $6.25A - A = 42$ or $5.25A = 42$ **[M1]**

8 **[A1]**

*[Total Marks: 9]*

**31 (a)** $\angle AXB = \angle CXD$ because they are vertically opposite angles **[B1]**

$\angle BAX = \angle DCX$ (or $\angle ABX = \angle CDX$) because they are alternate angles, $AB \parallel DC$ **[B1]**

All three pairs of corresponding angles are equal, so the triangles are similar (AAA) **[B1]**

**(b)** $\frac{2x-3}{x} = \frac{6}{4}$ or equivalent **[M1]**

$4(2x - 3) = 6x$ or $2x - 3 = 1.5x$ **[M1]**

$0.5x = 3$ leading to $x = 6$ **[A1]**

**(c)** Area of triangle $CDX = 24 \times 1.5^2$ or 54 **[M1]**

Recognises triangle $BCX$ shares altitude with $ABX$ and base ratio is 6 : 4 **[M1]**

Area of triangle $BCX$ (and triangle $ADX$) $= 24 \times 1.5$ or 36 **[M1]**

Total area $= 24 + 54 + 36 + 36 = 150$ **[A1]**

*[Total Marks: 10]*

**32 (a)** Finds the volume scale factor: $\frac{2700}{800} = 3.375$ (or $\frac{27}{8}$) **[M1]**

Finds the linear scale factor by cube rooting: $\sqrt[3]{3.375} = 1.5$ **[M1]**

$12 \times 1.5 = 18$ **[A1]**

**(b)** Finds the area scale factor from the linear scale factor: $1.5^2 = 2.25$ **[M1]**

Sets up the multiplication for the cost: $18 \times 2.25$ **[M1]**

40.50 **[A1]**

**(c)** Identifies the area scale factor between C and A is 1.44 **[B1]**

Finds the linear scale factor: $\sqrt{1.44} = 1.2$ **[M1]**

Finds the volume scale factor: $1.2^3 = 1.728$ **[M1]**

Calculates the volume of C: $800 \times 1.728 = 1382.4$ **[A1]**

*[Total Marks: 10]*

**33** **(a)** Slant height $l = \sqrt{10^2 + 24^2} = 26$ **[M1]**

Surface Area $= \pi(10)^2 + \pi(10)(26)$ **[M1]**

$360\pi$ **[A1]**

**(b)** $V_A = \frac{1}{3}\pi(10)^2(24) = 800\pi$ **[M1]**

Volume Ratio $= \frac{100\pi}{800\pi} = \frac{1}{8}$ **[M1]**

Area Ratio $= \left(\frac{1}{8}\right)^{\frac{2}{3}} = \frac{1}{4}$ **[M1]**

Surface Area of $B = 360\pi \times \frac{1}{4} = 90\pi$ **[A1]**

**(c)** Volume of frustum $= 800\pi - 100\pi = 700\pi$ **[M1]**

$\frac{4}{3}\pi R^3 = 700\pi$ **[M1]**

$R^3 = 700 \times \frac{3}{4} = 525$ **[M1]**

$R = \sqrt[3]{525} = 8.07$ (accept $8.066\ldots$) **[A1]**

*[Total Marks: 11]*